

Global CO₂ Emissions Analysis

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MSDS670 – Data Visualization

25-June-2026

Problem

CO2 emissions have increased by almost 200 times since 1850 yet the discussion about Who is responsible? is far from straightforward. When you consider emissions per person as well as per country, the responsibilities of the largest emitters (China, the USA) are very different from what is often conveyed in the headlines. In the absence of a clear, datadriven picture of trends, ranking and drivers of emissions, it is challenging for policymakers, business and the public to have an informed discussion on climate responsibility and priorities for mitigation. This project leverages a large, well documented global dataset in order to create that clearer picture.

Research Question

This project will answer three research questions:

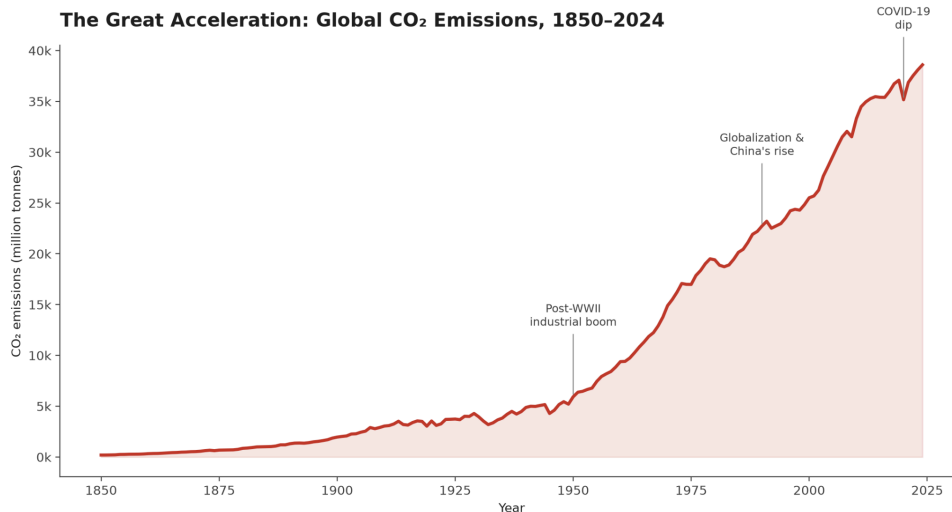
- **What has happened to CO2 emissions since 1850 and what do the trends indicate about industrialization and economic development?**
- **Which countries are the biggest emitters overall and which are the biggest per person and how much bigger or smaller are they?**
- **What are the strongest relationships between a country's emissions profile and its factors (GDP, energy use, population, fuel mix)?**

Methodology

These seven visualizations were created in Python, with pandas for data preparation, and matplotlib, seaborn, networkx and geopandas for plotting, and with the object oriented (stateless) matplotlib interface used for all of them. The type of chart was matched to the analytical purpose: line and stacked-area chart used to show trend over time; bar, scatter, map, heatmap, network – used to show cross-sectional snapshots of countries, compared one another.

Note on year selection for trend charts . They display all years available, as the point of a trend chart is to show change over time. A country may only be shown on one snapshot chart (bar, scatter, map, heatmap, network) at one time as there is only one reference year. The majority of snapshot charts are for the most recent year for which there is full country-level coverage (2023). Rather, the scatter chart and the correlation analysis are based on 2022; GDP is only available up to 2022 in this dataset, so 2023 and 2024 are not yet available. This information is made known in each chart's title to make it clear. All charts have a white background, no grid lines and limited, easily described legends.

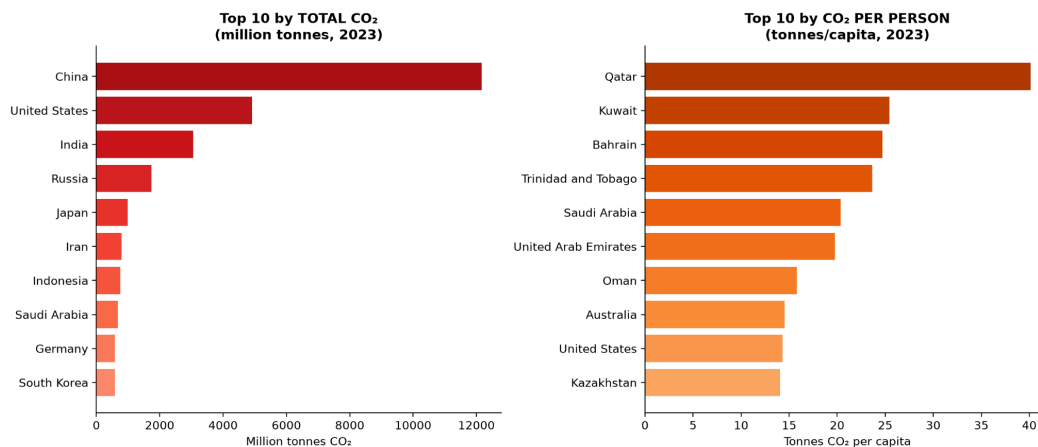
Results with Charts



CO₂ emissions for the globe between 1850 and 2024

Global trend: The amount of CO₂ released into the atmosphere has increased from a few hundred million tonnes in 1850 to about 38.6 billion tonnes in 2024. Three inflection points are evident: a second acceleration in the 1990s associated with the rise of globalization and the industrialization of China, and a sharp but temporary decline in 2020 brought on by COVID-19, with a rapid recovery to a new all-time high.

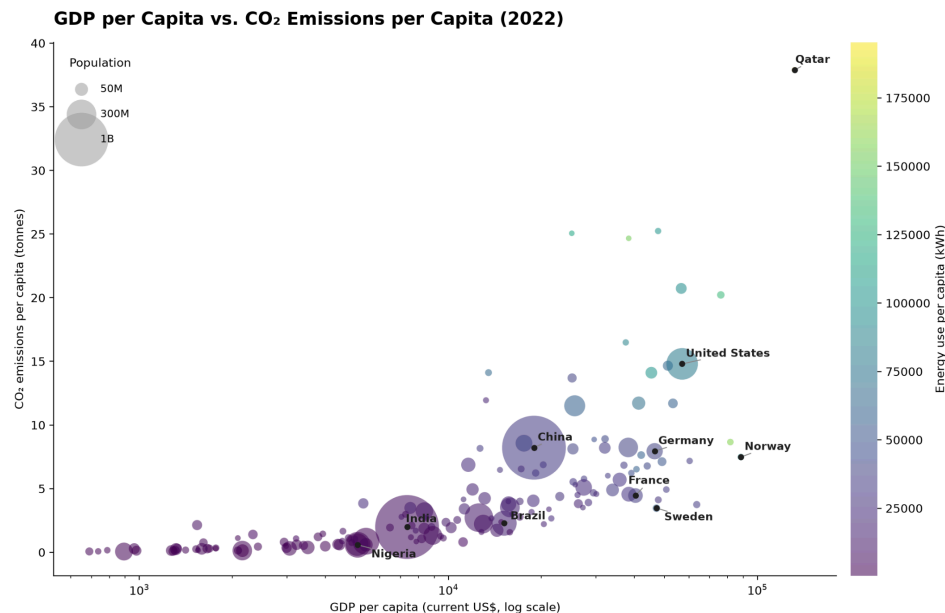
Top 10 Emitting Countries: Total CO₂ vs. CO₂ per Capita (2023)



Top 10 emitters in 2023 of global CO₂ emissions and CO₂ emissions per capita

Drivers of emissions: There is strong correlation between GDP per capita and CO₂ per capita (Spearman correlation of ~0.89) and between energy use per capita and CO₂

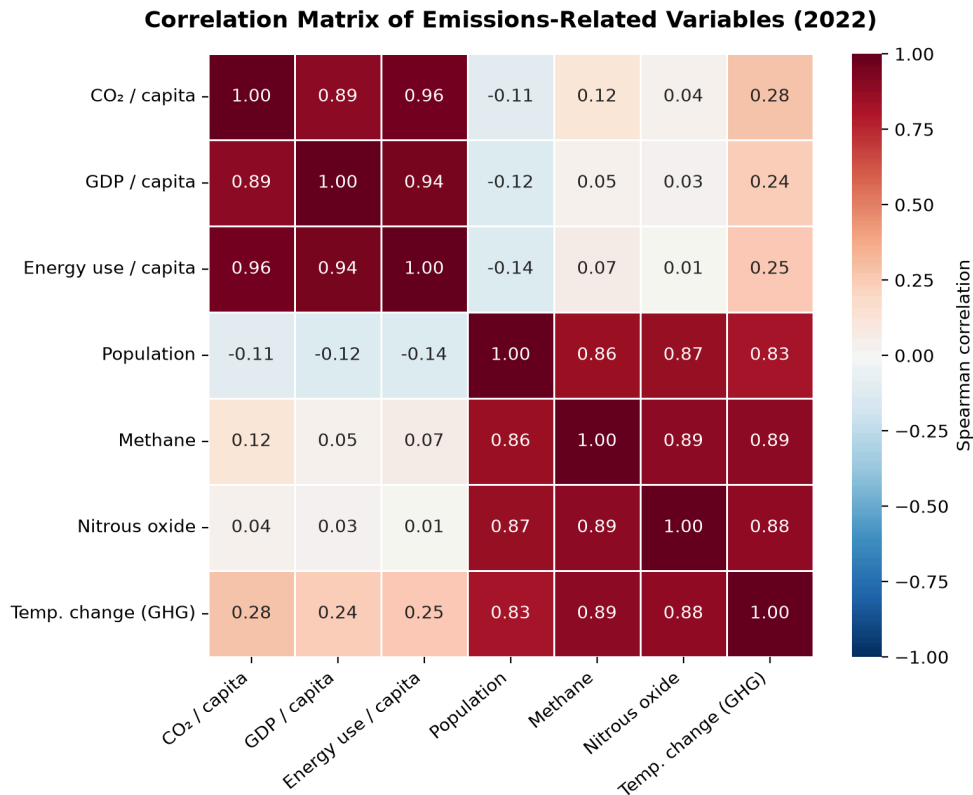
per capita (Spearman correlation of ~ 0.96), both of which indicate that wealth and energy use per capita are primary determinants of a country's per-person carbon footprint. The size of the population, on the other hand, exhibits virtually no correlation to per-capita emissions, but strong correlation to overall methane and nitrous oxide emissions. The scatter chart below reveals that there is not a direct link between wealth and CO₂ emissions, since Norway, Sweden and France have a lower emissions level per capita than other countries in a similar income range, suggesting that it is the energy mix that is different and not just wealth.



CO₂ emissions per capita vs. GDP per capita 2022

Geographic pattern: It is possible to map emissions per capita on a geographical basis, and this reveals that the highest per-capita emissions are found in North America, Australia and the Gulf states, with the majority of Africa and South Asia having much lower per-capita emissions. This pattern is more closely related to economic factors such as wealth and fossil-fuel export than geographic factors. The map shows the global distribution of CO₂ emissions in 2023 per capita.

Fuel mix and country clusters. The relationships between wealth and energy described above are confirmed by a broader array of variables in a correlation heatmap of emissions-related variables. A network graph also reveals that the top 30 emitting countries tend to be grouped into distinct clusters according to their fuel mix, with the Gulf oil exporters in one cluster, coal-dependent economies (Indonesia, Pakistan, Kazakhstan, Poland) in another, and diversified-fuel economies (Germany, Japan, UK, Canada) in a third, though somewhat less tightly connected cluster. Until the mid-20th century, coal was the most prevalent fuel used worldwide, with oil and gas becoming more and more important since the 1950s–60s and now accounting for the bulk of global emissions.



Emissions-related variables are correlated with each other for 2022

The correlations described above are confirmed and extended by the relations shown in the correlation heatmap of the seven country level variables: CO₂ per capita, GDP per capita, energy use per capita, population, methane, nitrous oxide, and temperature change attributable to GHGs. There is a strong correlation between CO₂ per capita and energy use per capita (0.96) and GDP per capita (0.89), meaning that energy use and GDP per capita (not population) are the main drivers of a country's individual carbon footprint. By contrast, CO₂ per capita has virtually no correlation with population, but strong correlations (0.83-0.87) with total methane, nitrous oxide, and temperature-change attribution — meaning that the more people that are present, the more total non-CO₂ GHGs are emitted even if each person's CO₂ emissions are low. The distinction is important for policy, as it is for reducing per-capita carbon intensity and reducing total volume of GHGs is a different lever entirely.

Geographical and income groupings of the top 30 emitting countries in coal, oil, gas and cement emissions show that countries are not grouped by income or geography, but by the type of emissions. The oil-exporting economies of the Gulf states (Saudi Arabia, Iraq, the United Arab Emirates) are grouped together in one fairly cohesive cluster, while the coal-dependent economies of Indonesia, Pakistan, Kazakhstan, and Poland make up a

second cluster, and diversified-fuel economies (Germany, Japan, the United Kingdom and Canada) form a third, somewhat less closely linked group. Clustering is not new, as coal, in the past, was the primary source of emissions until the 1950s and 1960s when oil and then mainly natural gas became the dominant fuels in the global mix. There's a structural gap between countries that haven't diversified their energy mix, and those that have, such that a single index based on wealth or geography wouldn't even register the fact that the latter are plotted further away from any particular cluster.

Key Conclusions

From this analysis there are three conclusions. First, global emissions growth has been non-stop since the beginning of industrialisation, with the only meaningful break being the temporary, externally driven drop during the COVID-19 pandemic and as yet no convincing indication of a longer-term structural decline. Second, who is responsible for emissions is entirely dependent on the metric used: total emissions, will go towards large, populous, industrializing economies, while per-capita emissions will go towards small, wealthy, fossil-fuel-intensive states. A one-sided narrative, whether it's a policy or a public narrative, is an incomplete story. Third, although wealth and its use of energy are the key trends in statistical explanations of country-level emissions intensity, it is clear that energy-mix policy, rather than just economic growth, is an important driver of the resulting emission trends meaning that the mix of fuels used in a country matters more than the energy consumed.

References

Our World in Data / Global Carbon Project. (2024). *CO2 and Greenhouse Gas Emissions Dataset*. <https://github.com/owid/co2-data>